means to mitigate even nonexistent threats, which would then feed the danger that deflection technology would be misused. For this reason, asteroid discovery and surveillance may not be a mere neutral tool of future policy, but rather a kind of booby trap. To me, the only foreseeable solution is a combination of accurate orbit estimation, realistic threat assessment, and effective public education—so that in democracies at least, the citizens can make their own, informed decisions. This is a job for NASA.

Near-Earth asteroids, and means of altering their orbits, are being looked at seriously. There is some sign that officials in the Department of Defense and the weapons laboratories are beginning to understand that there may be real dangers in planning to push asteroids around. Civilian and military scientists have met to discuss the subject. On first hearing about the asteroid hazard, many people think of it as a kind of Chicken Little fable; Goosey-Lucy, newly arrived and in great excitement, is communicating the urgent news that the sky is falling. The tendency to dismiss the prospect of any catastrophe that we have not personally witnessed is in the long run very foolish. But in this case it may be an ally of prudence.

MEANWHILE WE MUST STILL FACE the deflection dilemma. If we develop and deploy this technology, it may do us in. If we don't, some asteroid or comet may do us in. The resolution of the dilemma hinges, I think, on the fact that the likely timescales of the two dangers are very different—short for the former, long for the latter.

I like to think that our future involvement with near-Earth asteroids will go something like this: From ground-based observatories, we discover all the big ones, plot and monitor their orbits, determine rotation rates and compositions. Scientists are diligent in explaining the dangers—neither exaggerating nor muting the prospects. We send robotic spacecraft to fly by a few selected bodies, orbit them, land on them, and return surface samples to laboratories on Earth. Eventually we send humans. (Because of the low gravities, they will be able to make standing broad jumps of ten kilometers or more into the sky, and lob a baseball into orbit around